

Stat331 Assignment 3

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2026-07-28

Set up

Here are the libraries used:

```
library(data.table)
```

```
##
## Attaching package: 'data.table'

## The following object is masked from 'package:base':
##
##      %notin%
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::between()      masks data.table::between()
## x dplyr::filter()       masks stats::filter()
## x dplyr::first()        masks data.table::first()
## x lubridate::hour()     masks data.table::hour()
## x lubridate::isoweek()  masks data.table::isoweek()
## x lubridate::isoyear()  masks data.table::isoyear()
## x dplyr::lag()          masks stats::lag()
## x dplyr::last()         masks data.table::last()
## x lubridate::mday()     masks data.table::mday()
## x lubridate::minute()   masks data.table::minute()
## x lubridate::month()    masks data.table::month()
## x lubridate::quarter()  masks data.table::quarter()
## x lubridate::second()   masks data.table::second()
## x purrr::transpose()    masks data.table::transpose()
## x lubridate::wday()     masks data.table::wday()
## x lubridate::week()     masks data.table::week()
## x lubridate::yday()     masks data.table::yday()
## x lubridate::year()     masks data.table::year()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(leaps)
```

Here is also my A2 Question 2 code copy-pasted in, so that I can complete A3 without hassle:

```
#2.FROM A2
#a)
ceocomp <- read.csv('~/.Documents/GitHub/Big-Ahh-R-Repo/Stat331/Stat331 A2/data/raw/ceocomp.csv')
ceo_pos <- subset(ceocomp, ceocomp$PROF>0)
set.seed(21066922)
my.ceo_pos = ceo_pos[sample(nrow(ceo_pos), 60),]
my.ceo_pos
```

##	COMP	AGE	EDUCATN	BACKGRD	TENURE	EXPER	SALES	VAL	PCNTOWN	PROF
## 50	616	51	2	3	16	9.0	2754	0.3	0.01	237
## 31	1331	56	2	2	33	4.0	18807	4.8	0.03	942
## 67	925	64	1	4	39	30.0	3670	46.5	2.52	118
## 85	1032	63	2	5	35	8.0	2354	2.2	0.12	46
## 6	2342	71	1	1	46	34.0	835	9.6	0.83	99
## 41	1948	55	1	3	23	23.0	1227	7.6	0.55	145
## 25	768	60	2	1	30	8.0	2097	0.7	0.02	322
## 94	854	62	1	5	30	3.0	669	2.6	0.17	81
## 27	307	47	2	1	14	2.0	649	0.2	0.02	65
## 57	466	48	2	3	17	1.0	1539	0.2	0.01	189
## 95	600	60	0	5	42	1.0	1214	85.6	17.66	43
## 68	596	60	2	4	9	2.0	1148	0.3	0.02	136
## 64	558	62	2	4	34	12.0	2359	0.2	0.01	265
## 49	1031	55	1	3	35	1.0	13340	3.6	0.02	1154
## 74	2094	63	1	5	41	8.0	10818	5.9	0.04	1166
## 12	1030	55	2	1	11	11.0	1236	3.3	0.43	22
## 87	1019	56	2	5	30	9.0	3259	49.3	0.68	311
## 89	600	52	2	5	12	2.0	21351	4.2	0.26	101
## 59	597	57	2	4	27	13.0	2686	1.4	0.03	377
## 32	1034	51	1	2	23	3.0	578	7.0	0.83	82
## 47	1652	60	2	3	34	14.0	14949	11.5	0.06	856
## 16	447	50	2	1	5	1.0	1084	0.3	0.03	54
## 1	570	60	1	1	3	3.0	804	0.2	0.21	49
## 39	517	59	1	2	37	21.0	3180	181.0	6.70	136
## 88	1752	63	1	5	14	14.0	5137	534.2	3.99	249
## 71	809	59	1	5	38	0.5	19196	0.4	0.01	505
## 11	514	57	2	1	3	3.0	661	4.1	0.17	79
## 35	1571	53	1	2	31	9.0	16115	4.2	0.09	301
## 63	2244	64	2	4	41	5.0	4451	4.0	0.04	30
## 79	453	64	2	5	40	26.0	872	54.5	3.14	91
## 62	670	64	2	4	15	15.0	1535	1.9	0.12	145
## 72	2027	62	2	5	25	5.0	8379	3.4	0.04	806
## 76	1126	64	1	5	28	14.0	2545	3.4	0.21	63
## 14	576	48	2	1	8	6.0	1199	0.3	0.02	134
## 70	804	55	0	5	15	10.0	657	10.6	1.75	25
## 30	541	51	2	2	17	12.0	862	1.0	0.10	113
## 9	1341	55	2	1	30	7.0	4377	2.1	0.04	390
## 13	329	37	2	1	11	1.0	344	12.9	4.15	29
## 46	635	59	2	3	33	2.0	3963	0.8	0.12	99
## 96	1088	45	2	5	11	11.0	415	2.6	0.17	48
## 17	1144	60	2	1	37	10.0	4594	6.9	0.37	150
## 44	2847	62	1	3	37	10.0	3817	3.6	0.05	584
## 80	643	65	1	5	29	23.0	1681	8.9	0.89	44
## 48	467	62	2	3	13	8.0	2549	0.3	0.01	306

```
## 61 893 60      2      4      36 15.0 4444      2.0      0.04 417
## 52 1377 56      1      3      32  3.0 2434      5.3      0.18 273
## 84 493 59      1      5      34  4.0 1556      0.2      0.01 169
## 38 1622 57      1      2      29  5.0 9886      9.7      0.19 347
## 53 621 60      2      3      36  3.0 1552      0.9      0.03 210
## 20 688 54      2      1      28 28.0 3277 1689.0 34.04 317
## 36 1554 61      1      2      41 17.0 1435      30.6      2.23  82
## 37 427 54      1      2      31 15.0 2366      71.4     10.09  55
## 75 1833 63      2      5      36  7.0 17037      6.3      0.04 709
## 91 1630 60      1      5      36 20.0 8331     132.5      8.21  82
## 23 476 50      1      1      20  0.5 3148      1.3      0.05 260
## 45 897 56      2      3       2  5.0 2689      1.0      0.07  81
## 33 1461 63      1      2      23  8.0 5673      3.6      0.05 201
## 69 462 61      2      4      25 11.0 1314      16.8      1.03  27
## 18 425 74      1      1       3  3.0  475      0.1      0.07   6
## 19 1671 63      2      1      17 12.0 1967      5.5      0.20 143
```

```
comp <- my.ceo_pos$COMP
x1 <- my.ceo_pos$AGE
x2 <- my.ceo_pos$EDUCATN
x3 <- as.factor(my.ceo_pos$BACKGRD)
x4 <- my.ceo_pos$TENURE
x5 <- my.ceo_pos$EXPER
x6 <- my.ceo_pos$SALES
x7 <- my.ceo_pos$VAL
x8 <- my.ceo_pos$PCNTOWN
x9 <- my.ceo_pos$PROF
fullmodel <- lm(comp ~ x1+x2+x3+x4+x5+x6+x7+x8+x9)
summary(fullmodel)
```

```
##
## Call:
## lm(formula = comp ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -770.55 -243.91  -58.34   244.04 1377.97
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   47.44104   813.18192    0.058   0.9537
## x1             12.97890    13.09608    0.991   0.3267
## x2            -179.31590   142.78847   -1.256   0.2154
## x32            -56.74491   245.13235   -0.231   0.8179
## x33             -2.10798   213.67373   -0.010   0.9922
## x34            -289.68693   238.66829   -1.214   0.2309
## x35            -106.44351   205.82659   -0.517   0.6075
## x4              11.83513     7.80118    1.517   0.1359
## x5              16.16317    10.72494    1.507   0.1385
## x6               0.01224     0.01796    0.682   0.4989
## x7               0.82042     0.66549    1.233   0.2238
## x8            -58.36783    29.68720   -1.966   0.0552
## x9               0.49202     0.36009    1.366   0.1783
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 500.2 on 47 degrees of freedom
## Multiple R-squared:  0.4259, Adjusted R-squared:  0.2793
## F-statistic: 2.905 on 12 and 47 DF,  p-value: 0.004416

#b)
#keep in mind the hypothesis test is for betahat = 0 vector
r2_full <- summary(fullmodel)$r.squared
msreg <- r2_full/12
msres <- (1-r2_full)/47
1 - pf(msreg/msres,12,47)

## [1] 0.004415791

#c)
#i) with beta hat x34 being -289, we say WITHOUT statistical significance that lawyers tend to make les
#ii) sales also makes less, but we're even LESS certain of this estimate.
#iii)
redmodel <- lm(comp ~ x1+x2+x4+x5+x6+x7+x8+x9)
r2_red <- summary(redmodel)$r.squared

f_num <- (r2_full - r2_red)/4
f_den <- (1 - r2_full)/47
f_stat <- f_num/f_den
1 - pf(f_stat,4,47)

## [1] 0.7764592

#iv)
anova(redmodel,fullmodel)

## Analysis of Variance Table
##
## Model 1: comp ~ x1 + x2 + x4 + x5 + x6 + x7 + x8 + x9
## Model 2: comp ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      51 12201235
## 2      47 11757365   4   443870 0.4436 0.7765

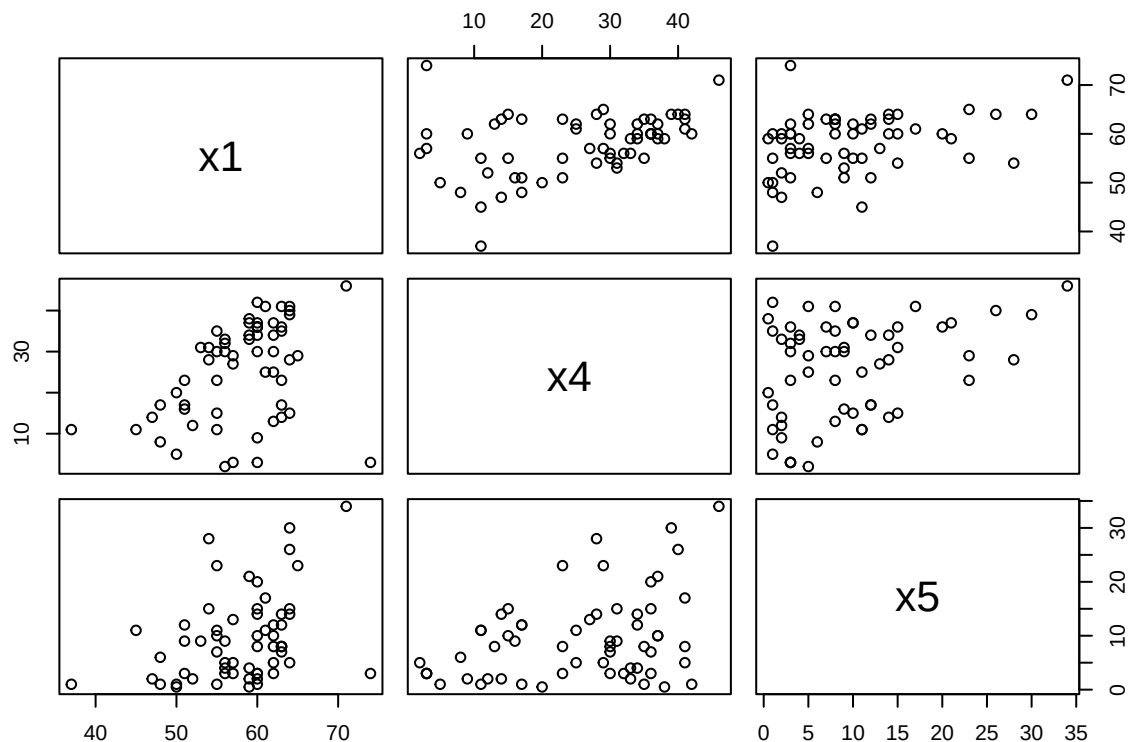
#d)
summary(redmodel)

##
## Call:
## lm(formula = comp ~ x1 + x2 + x4 + x5 + x6 + x7 + x8 + x9)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -717.60 -285.72  -39.28   286.79 1311.98
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  277.33972   730.43905   0.380   0.7058
## x1             9.17031    12.13919   0.755   0.4535
## x2          -212.08401    125.88116  -1.685   0.0981 .
## x4             10.44381     7.40399   1.411   0.1644
## x5             15.04509    10.37970   1.449   0.1533
```

```
## x6          0.01059    0.01619    0.654    0.5158
## x7          0.88585    0.64852    1.366    0.1779
## x8         -59.34763    28.94672   -2.050    0.0455 *
## x9          0.55425    0.33688    1.645    0.1061
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 489.1 on 51 degrees of freedom
## Multiple R-squared:  0.4042, Adjusted R-squared:  0.3108
## F-statistic: 4.325 on 8 and 51 DF,  p-value: 0.0004988
```

#i) they are all correlated, as a long EXPER certainly implies longer TENURE and AGE
#ii)

```
pairs(cbind(x1,x4,x5))
```



#iii)

```
agemodel <- lm(x1~x2+x4+x5+x6+x7+x8+x9)
summary(agemodel)
```

```
##
## Call:
## lm(formula = x1 ~ x2 + x4 + x5 + x6 + x7 + x8 + x9)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.0663  -2.7035   0.0506   2.9182  19.2603
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.589e+01  3.090e+00  18.089  <2e-16 ***
## x2          -2.245e+00  1.404e+00  -1.599   0.1159
```

```
## x4          1.743e-01  8.106e-02  2.150  0.0362 *
## x5          2.137e-01  1.148e-01  1.861  0.0684 .
## x6         -6.308e-05  1.847e-04 -0.341  0.7341
## x7          7.964e-03  7.326e-03  1.087  0.2820
## x8         -6.128e-01  3.196e-01 -1.918  0.0607 .
## x9         -2.355e-04  3.848e-03 -0.061  0.9514
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.588 on 52 degrees of freedom
## Multiple R-squared:  0.2956, Adjusted R-squared:  0.2008
## F-statistic: 3.118 on 7 and 52 DF,  p-value: 0.007999

VIF <- 1/(1-0.2956)
#its ok
#e)
new_x <- data.frame(x1=48,x2=1,x4=14,x5=3,x6=3100,x7=2.1,x8=1.1,x9=172)
predict(redmodel,new_x,interval='prediction',level=.95)

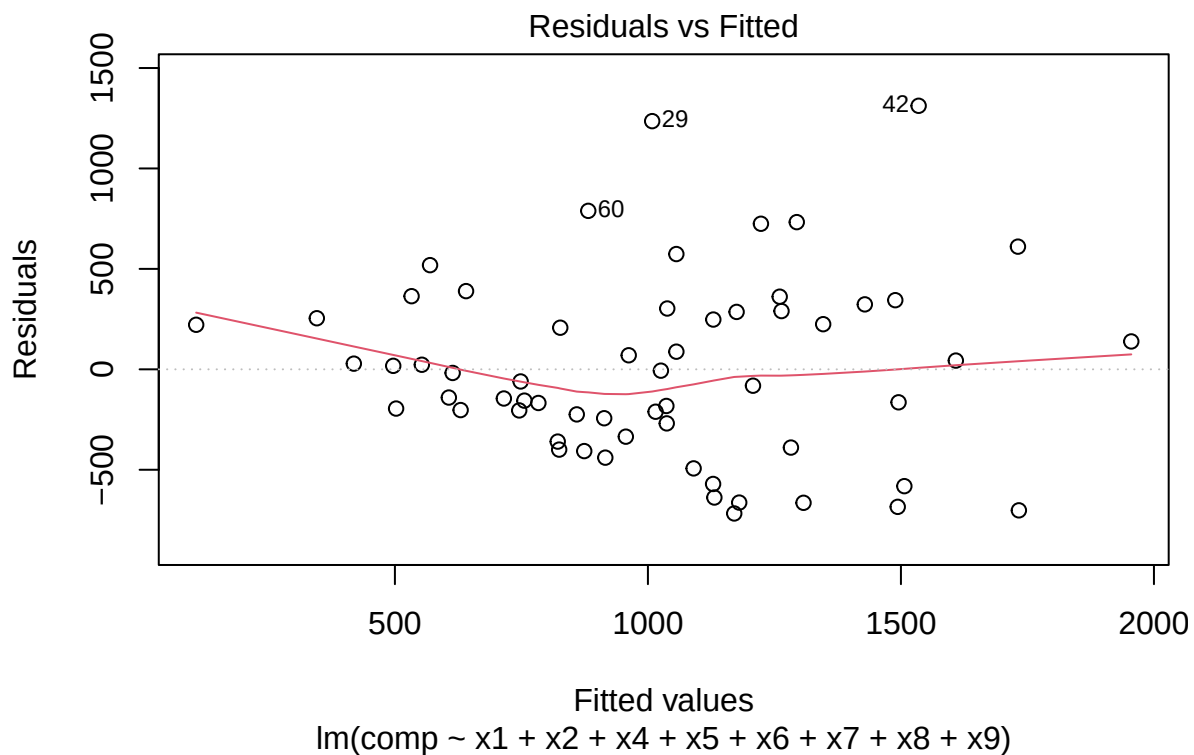
##          fit          lwr          upr
## 1 761.5257 -268.8854 1791.937
```

FROM HERE ON IS THE START OF A3

Question 1

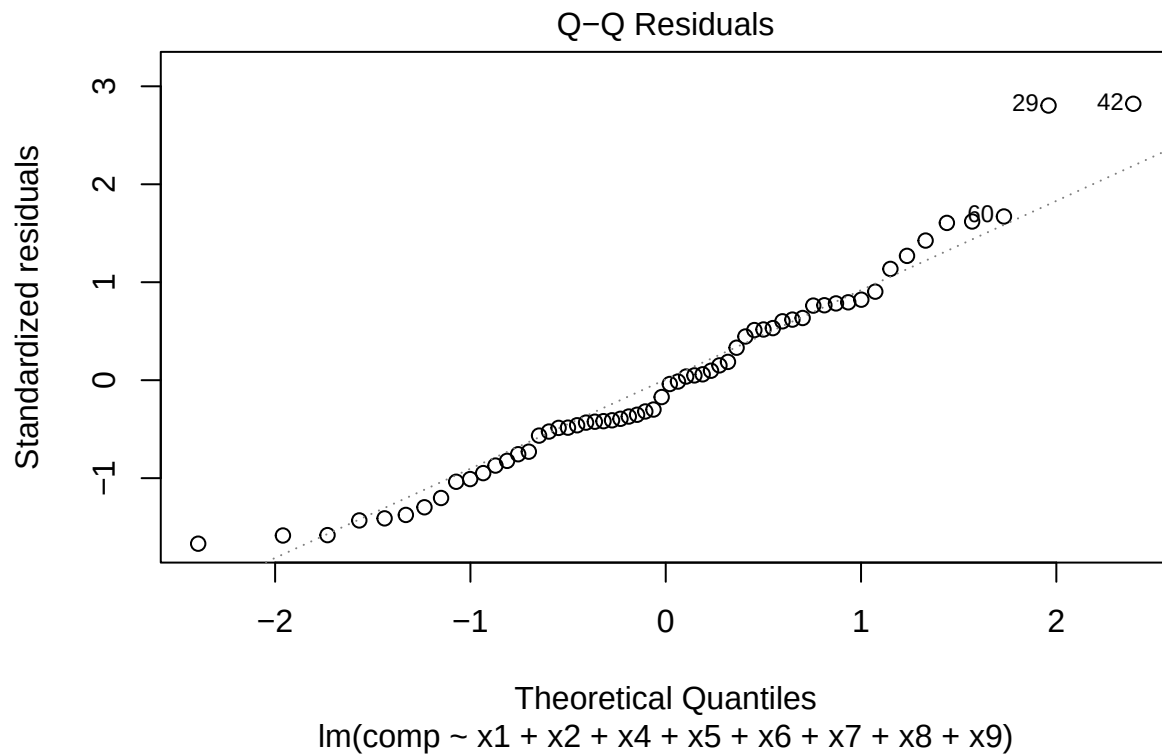
The residual vs fitted plot is fairly regular, nothing too out of the blue. Some outliers, but that's to be expected.

```
plot(redmodel, which = 1)
```



The QQ is also rather normal, hinting at a slight right skew, but overall I think it's meets expectations of being roughly normally distributed.

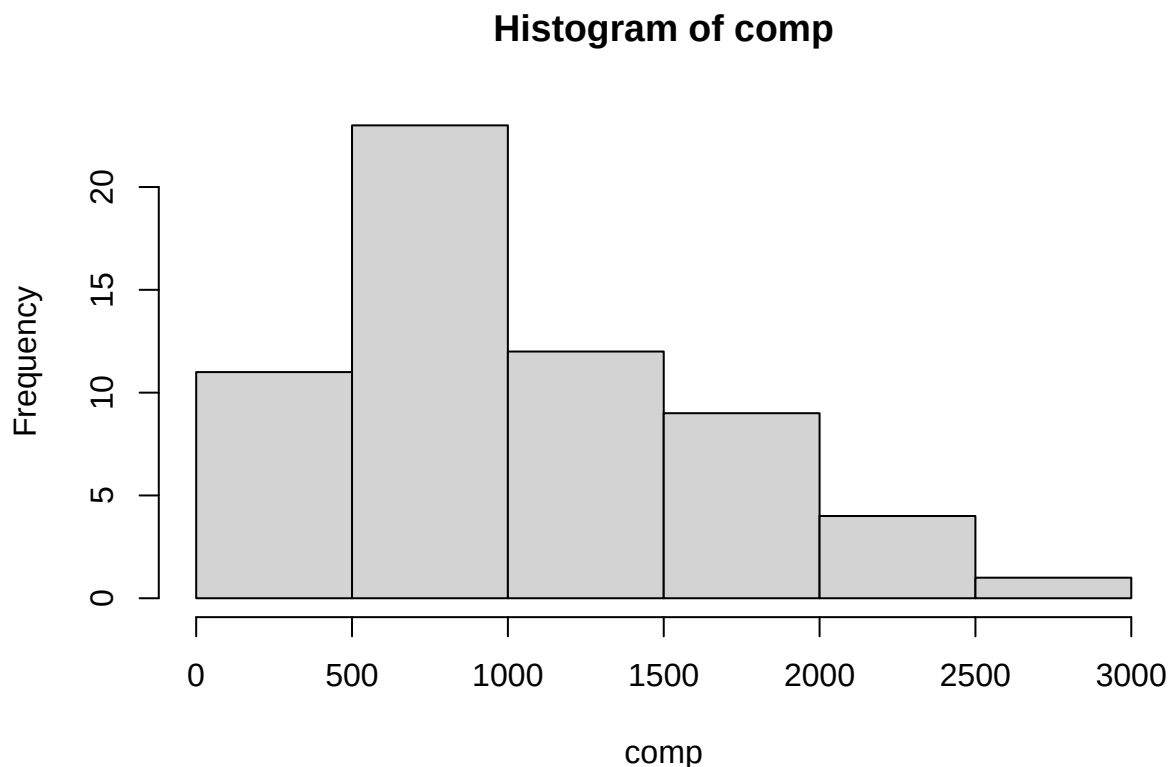
```
plot(redmodel, which = 2)
```



Question 2

a)

```
hist(comp)
```



it has a heavy right tail, so using $\ln$ as a transform would send the outliers a lot closer to the rest of the data.

b)

```
lncomp <- log(comp)
lnredmodel <- lm(lncomp ~ x1+x2+x4+x5+x6+x7+x8+x9)
```

c)

```
summary(lnredmodel)
```

```
##
## Call:
## lm(formula = lncomp ~ x1 + x2 + x4 + x5 + x6 + x7 + x8 + x9)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.78408	-0.28263	-0.03879	0.33245	0.91600

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.056e+00	6.680e-01	9.066	3.27e-12 ***
x1	8.821e-03	1.110e-02	0.795	0.4305
x2	-2.093e-01	1.151e-01	-1.818	0.0750 .
x4	1.066e-02	6.771e-03	1.574	0.1218
x5	1.384e-02	9.493e-03	1.458	0.1509
x6	1.744e-05	1.481e-05	1.178	0.2442
x7	1.013e-03	5.931e-04	1.708	0.0936 .

```
## x8          -6.259e-02  2.647e-02  -2.364   0.0219 *
## x9          4.131e-04  3.081e-04   1.341   0.1859
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4473 on 51 degrees of freedom
## Multiple R-squared:  0.4391, Adjusted R-squared:  0.3511
## F-statistic: 4.991 on 8 and 51 DF,  p-value: 0.000134
```

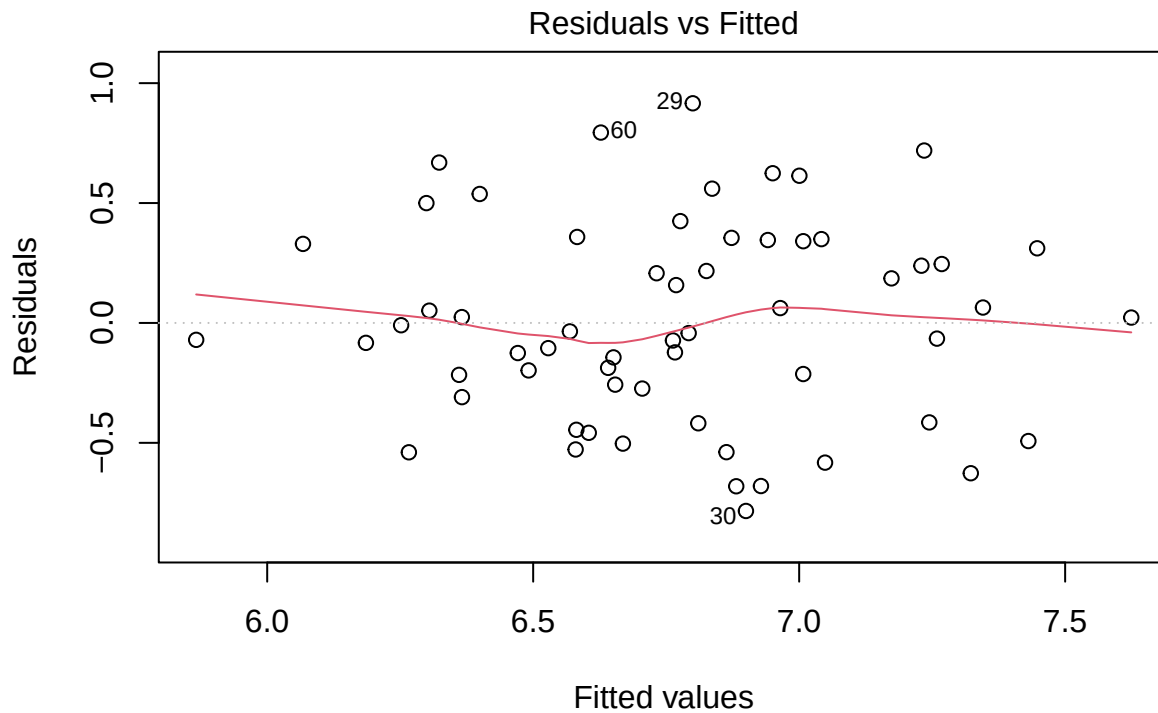
```
summary(redmodel)
```

```
##
## Call:
## lm(formula = comp ~ x1 + x2 + x4 + x5 + x6 + x7 + x8 + x9)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -717.60 -285.72  -39.28   286.79 1311.98
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   277.33972   730.43905    0.380   0.7058
## x1              9.17031    12.13919    0.755   0.4535
## x2            -212.08401    125.88116   -1.685   0.0981 .
## x4              10.44381     7.40399    1.411   0.1644
## x5              15.04509    10.37970    1.449   0.1533
## x6               0.01059     0.01619    0.654   0.5158
## x7               0.88585     0.64852    1.366   0.1779
## x8            -59.34763    28.94672   -2.050   0.0455 *
## x9               0.55425     0.33688    1.645   0.1061
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 489.1 on 51 degrees of freedom
## Multiple R-squared:  0.4042, Adjusted R-squared:  0.3108
## F-statistic: 4.325 on 8 and 51 DF,  p-value: 0.0004988
```

we see a very slight improvement in the model in which we applied log to the compensation.

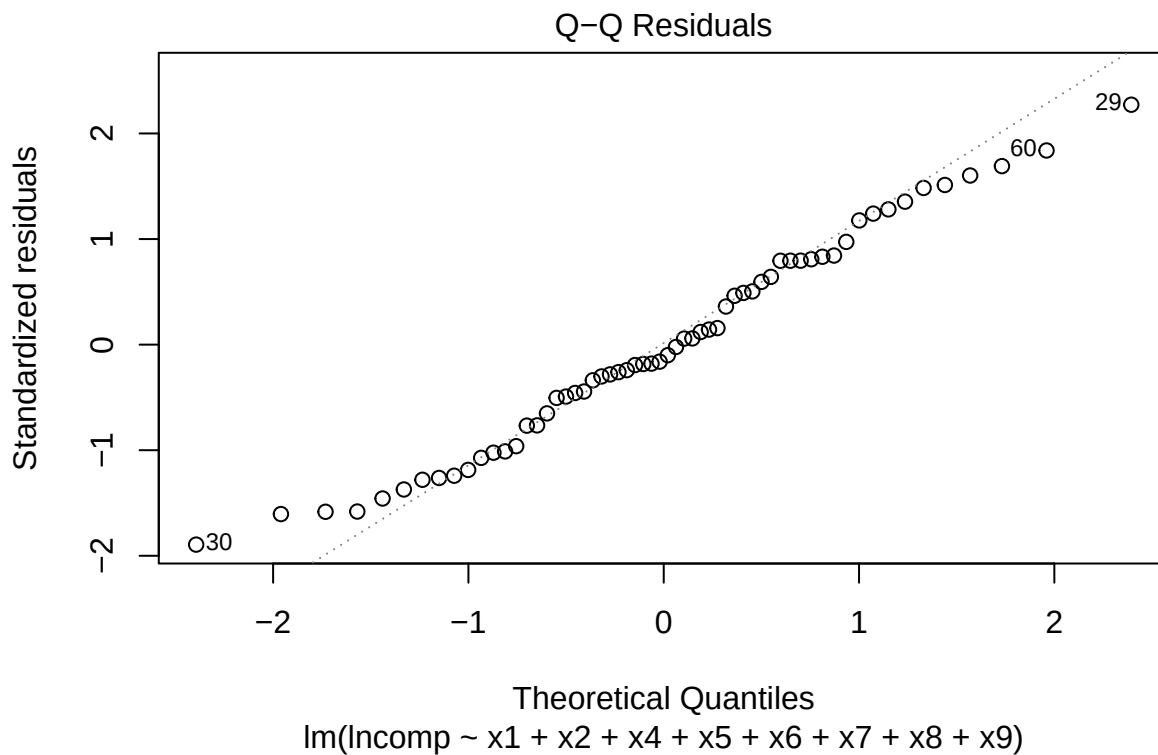
d)

```
plot(lnredmodel, which = 1)
```



I personally don't see a substantial difference in the res vs the fitted, but that's about normal for my human eyes.

```
plot(lnredmodel, which = 2)
```

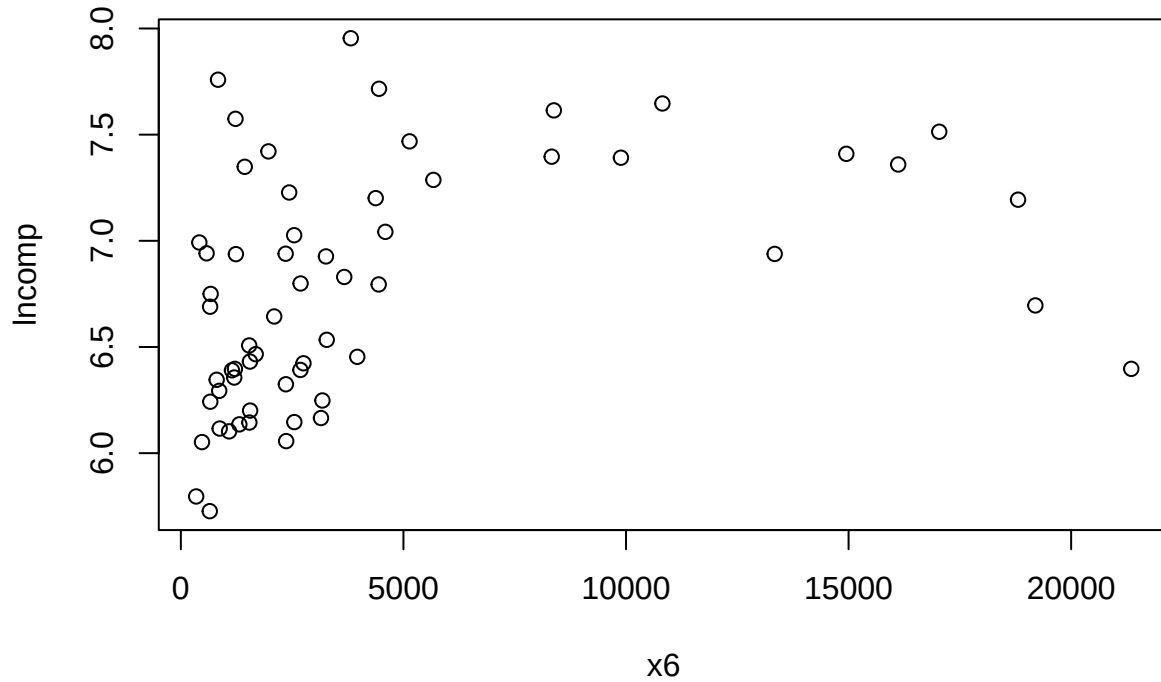


Here I notice that the skew is gone from before, but there is slightly lighter tails than expected of a normal distribution.

Question 3

a)

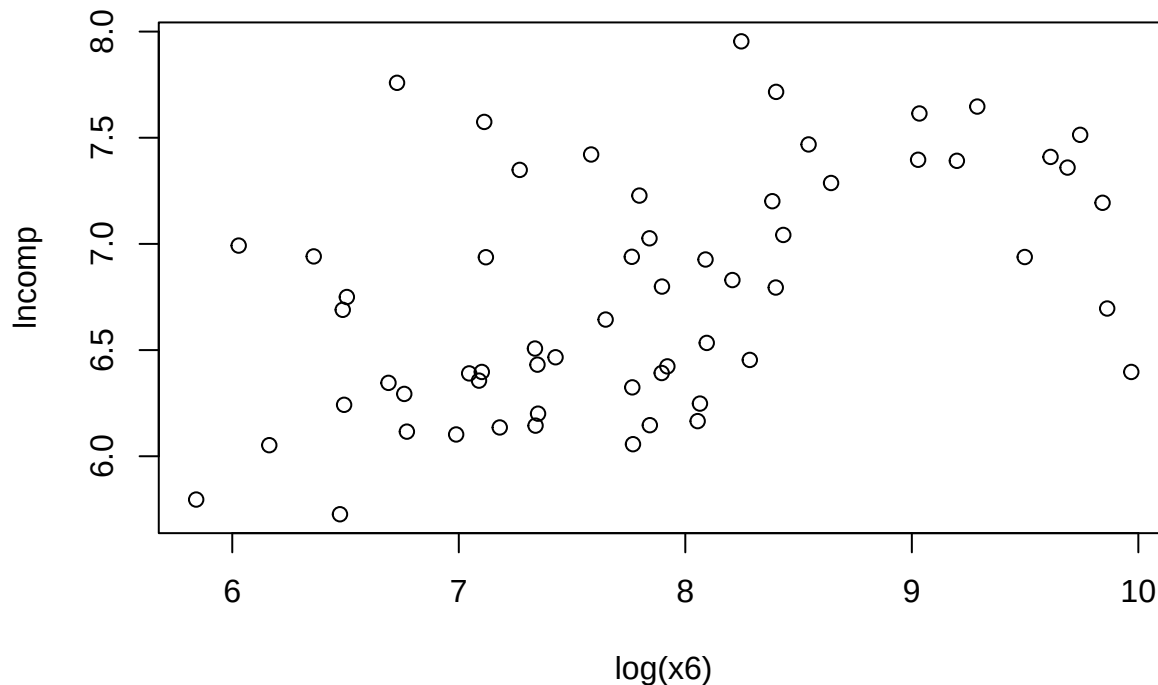
```
plot(x6,lncomp)
```



Given the sales and log of compensation, it doesn't really seem like a linear model would be a good fit, the data doesn't seem to have a strong linear relation.

b)

```
plot(log(x6),lncomp)
```



We see a slightly better linear relation, but it's still not particularly clear, especially on the far right side.

c)

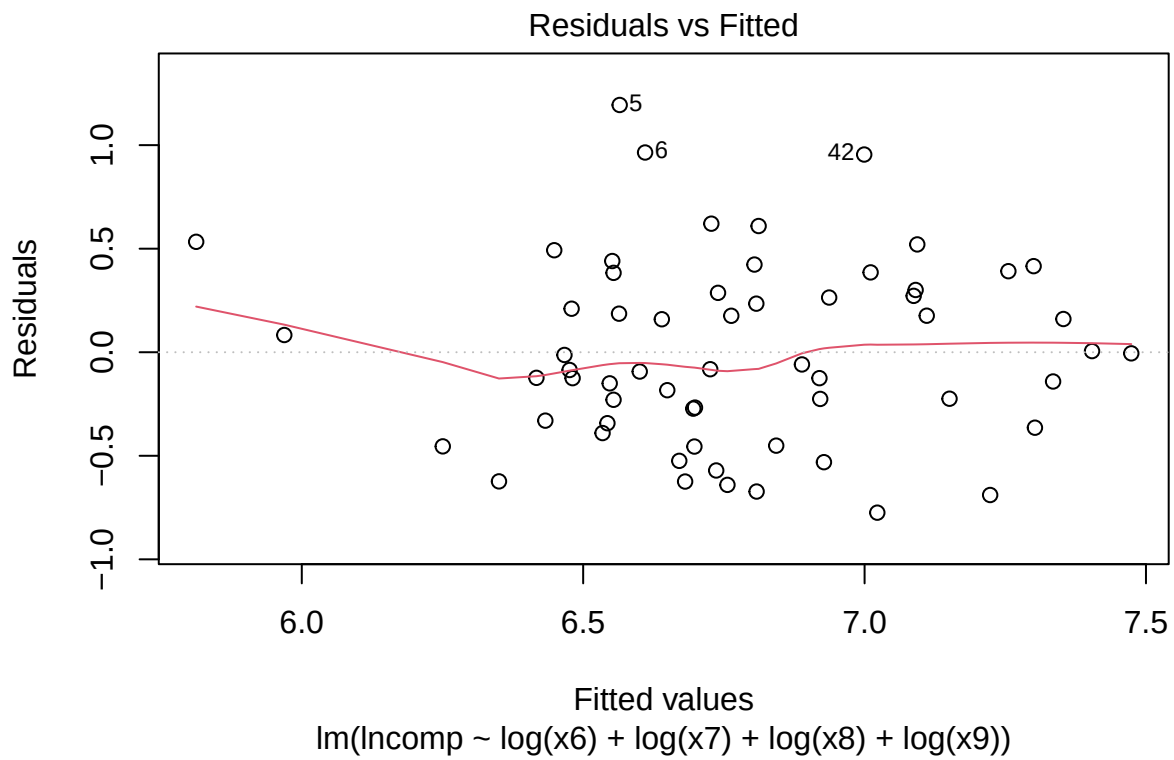
```
lnmodel <- lm(lncomp ~ log(x6)+log(x7)+log(x8)+log(x9))
summary(lnmodel)
```

```
##
## Call:
## lm(formula = lncomp ~ log(x6) + log(x7) + log(x8) + log(x9))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.77458 -0.33331 -0.07043  0.29030  1.19390
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.49990    0.51439  10.692 4.81e-15 ***
## log(x6)        0.09903    0.08656   1.144  0.25755
## log(x7)        0.29297    0.10152   2.886  0.00557 **
## log(x8)       -0.24594    0.10592  -2.322  0.02397 *
## log(x9)       -0.06740    0.10216  -0.660  0.51217
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4585 on 55 degrees of freedom
## Multiple R-squared:  0.3644, Adjusted R-squared:  0.3182
## F-statistic: 7.884 on 4 and 55 DF,  p-value: 4.261e-05
```

We can notice that the log of x7(Val), and log of x8(PCTOWN) have p values of significance, so we expect that these would explain more of the variation than the other 2 variates. Overall, it is a weaker model, with a smaller adj R^2 than the model previously, which only took the log of the response variate.

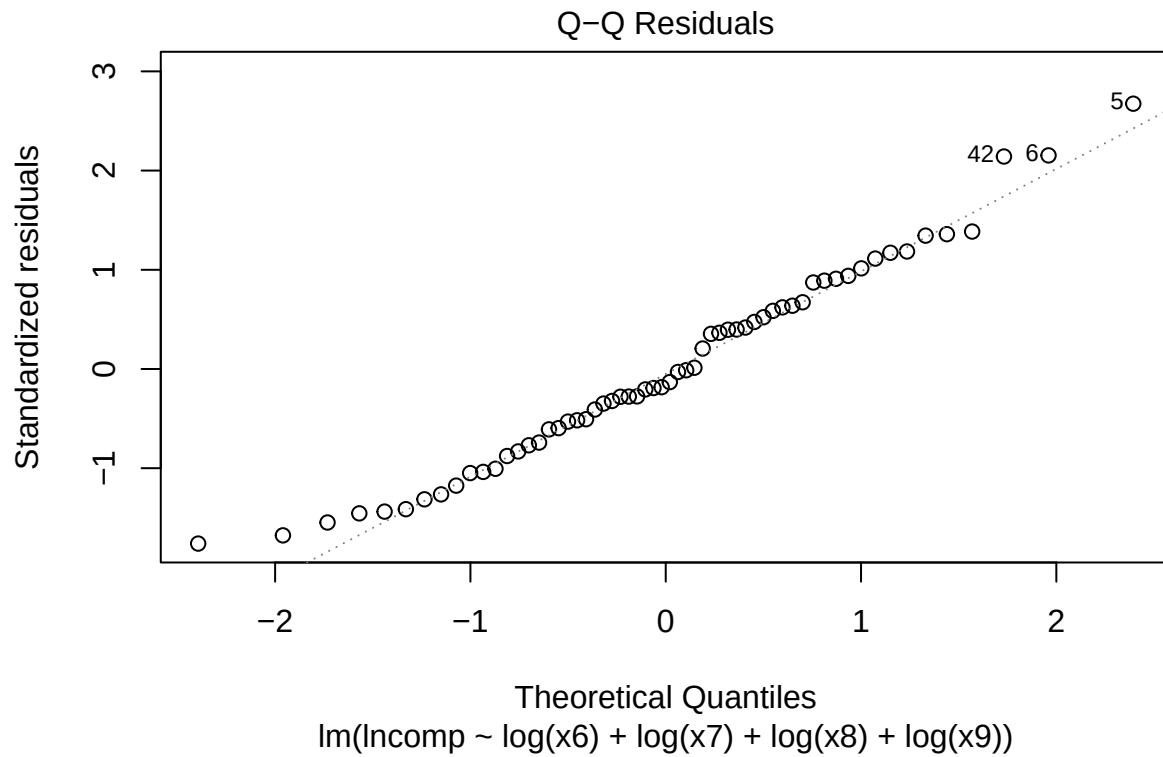
Question 4

```
plot(lnmodel, which = 1)
```



It is getting marginally better, with a flatter line, concentrating at 0.

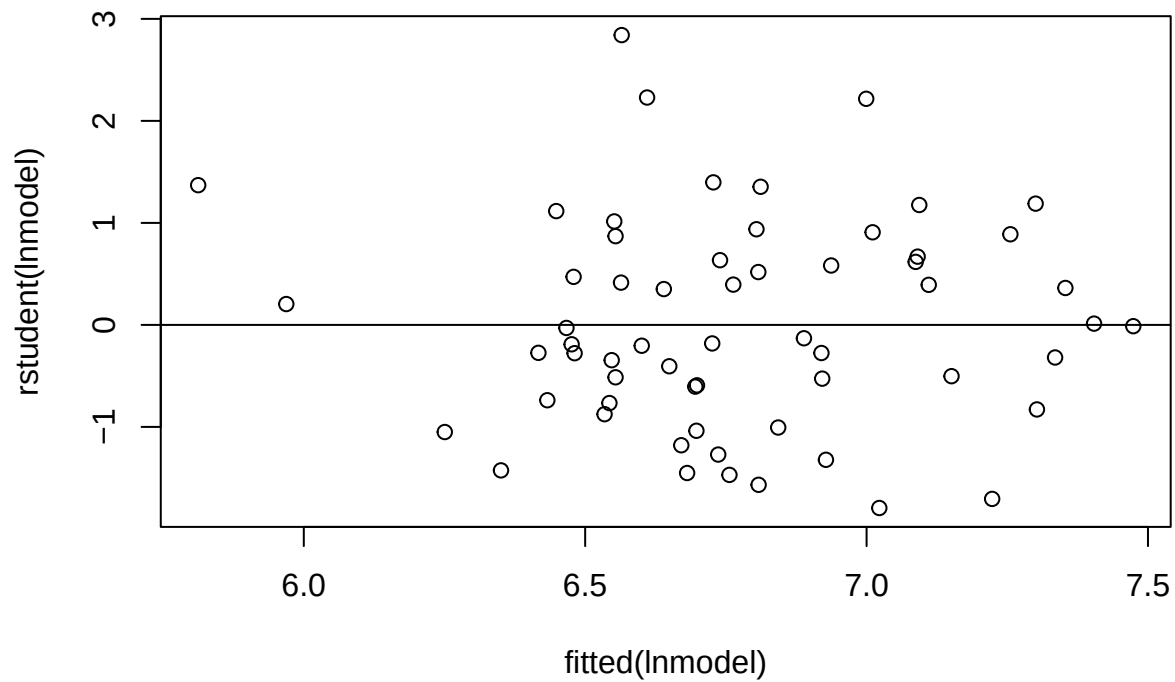
```
plot(lnmodel, which = 2)
```



The

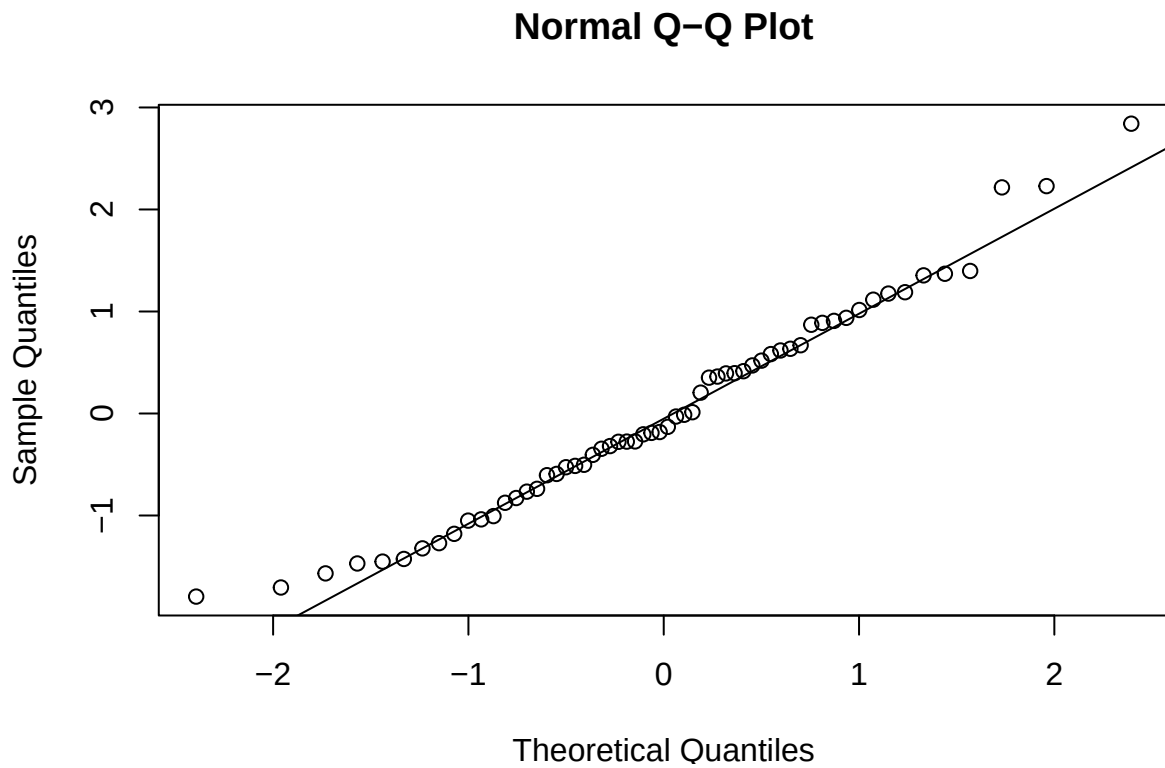
Question 5

```
plot(fitted(lnmodel), rstudent(lnmodel))
abline(h=0)
```



We see that it is really similar to the previous plot.

```
qqnorm(rstudent(lnmodel)); qqline(rstudent(lnmodel))
```



Same

here, it is rather similar.

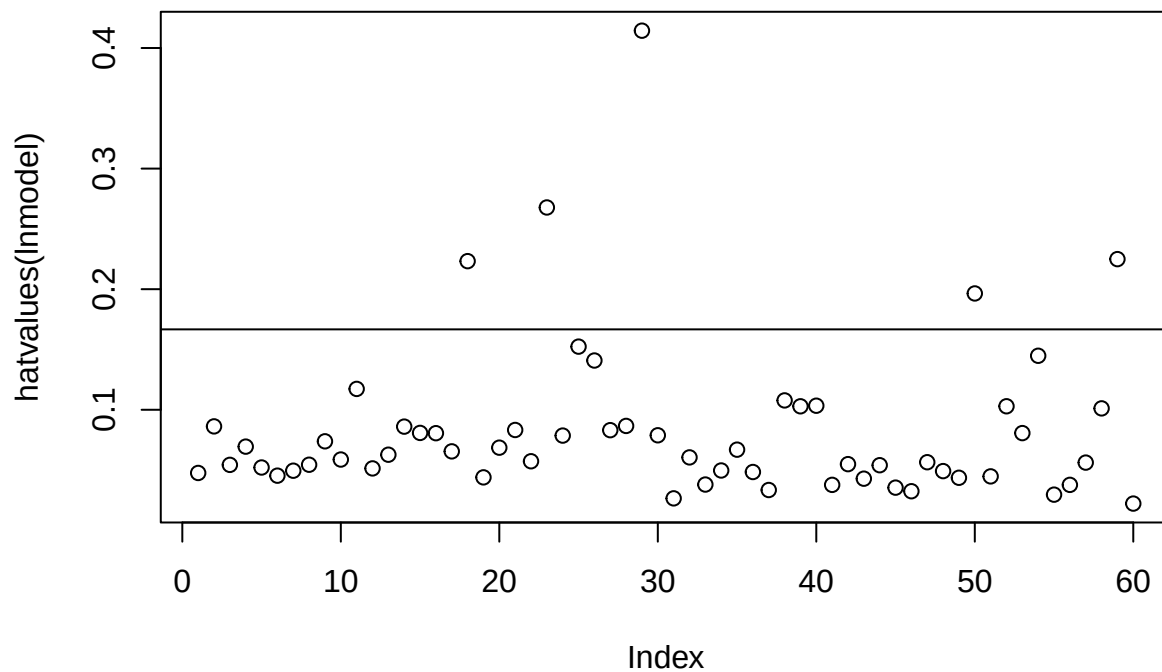
Question 6

remember that $p = 4$, $n = 60$, so we consider 'high-leverage' to be:

```
highlev = 2*(4+1)/60  
highlev
```

```
## [1] 0.1666667
```

```
plot(hatvalues(lnmodel))  
abline(h = highlev)
```



there is indeed a big outlier. Let's investigate.

We see

Question 7

Here is the highest leverage index:

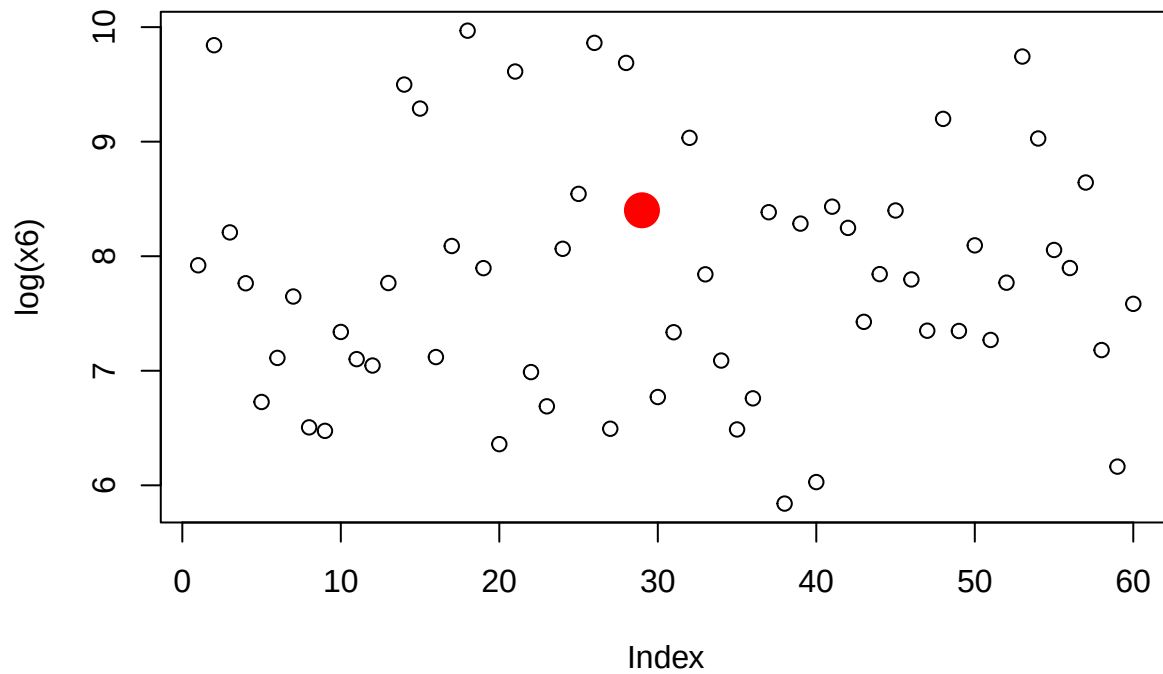
```
which.max(hatvalues(lnmodel))
```

```
## 29
```

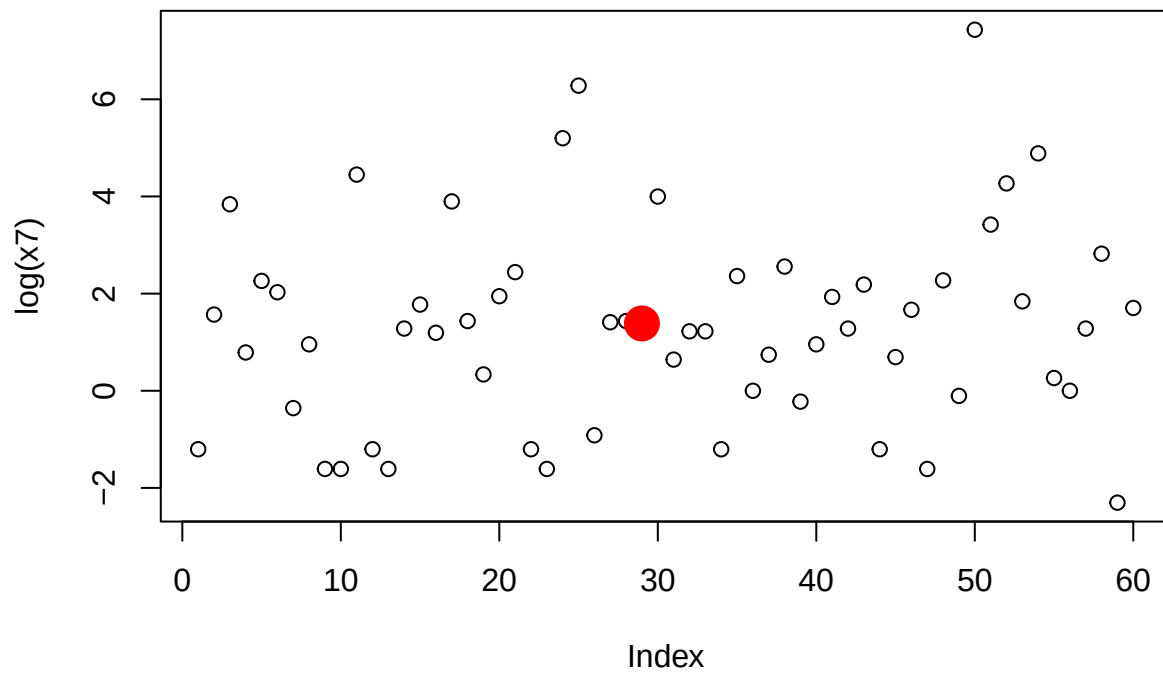
```
## 29
```

Here are the plots of the explanatory variates with the 29th index highlighted:

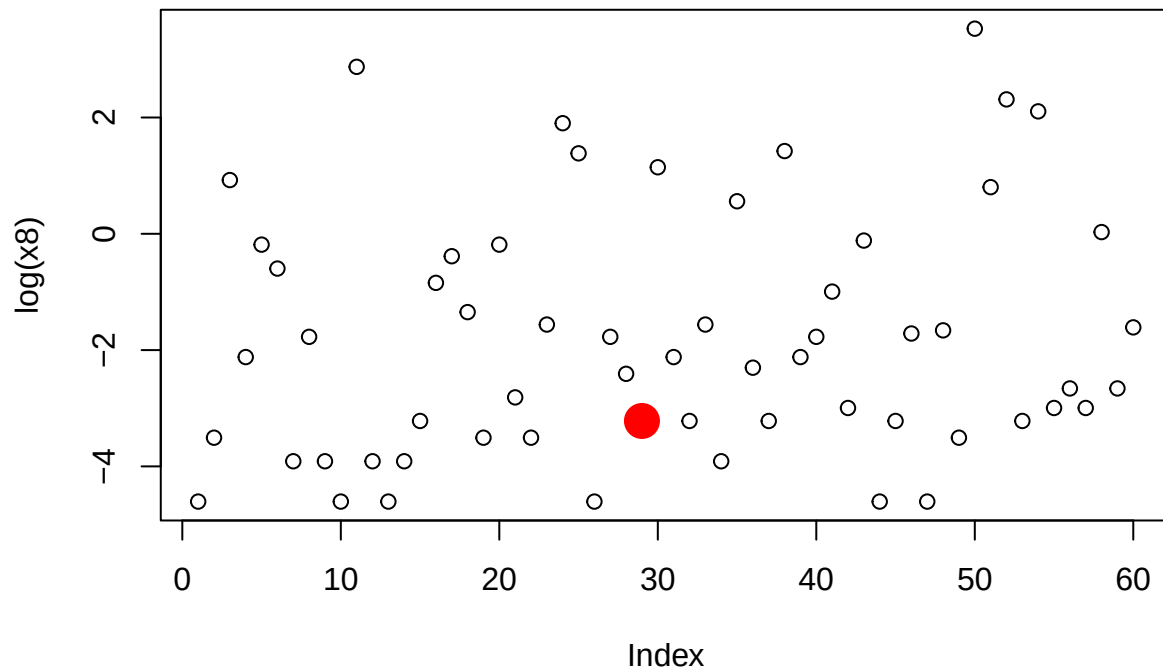
```
plot(log(x6))
points(29, log(x6)[29], col = "red", pch = 16, cex = 2.5)
```



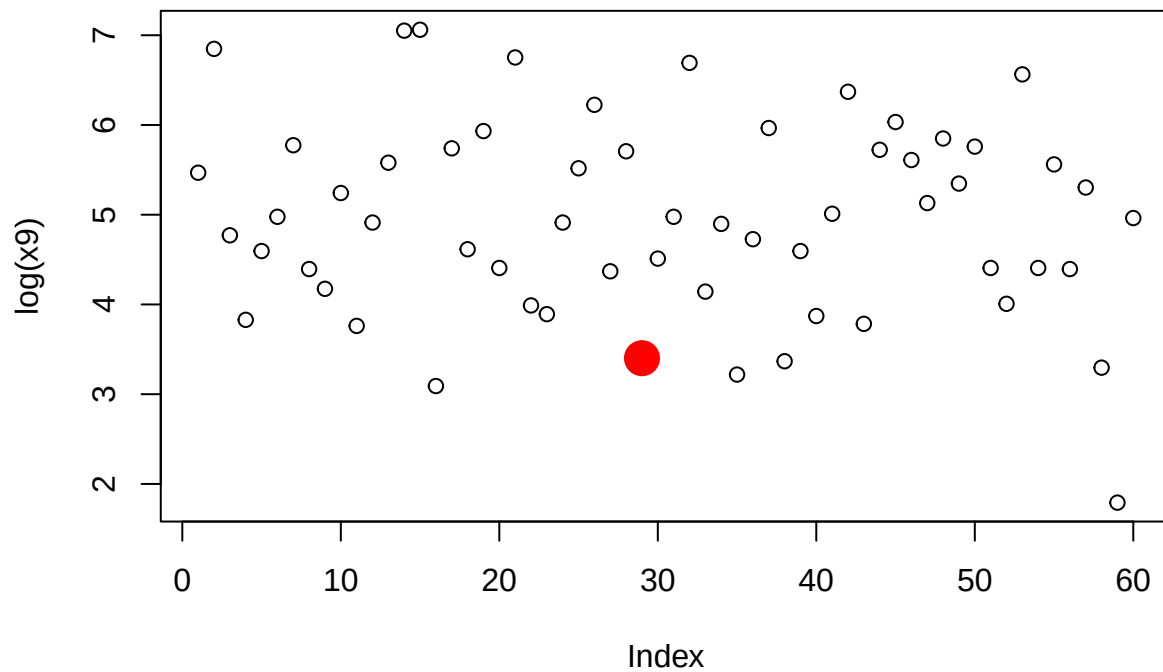
```
plot(log(x7))
points(29,log(x7)[29], col = "red", pch = 16, cex = 2.5)
```



```
plot(log(x8))
points(29,log(x8)[29], col = "red", pch = 16, cex = 2.5)
```



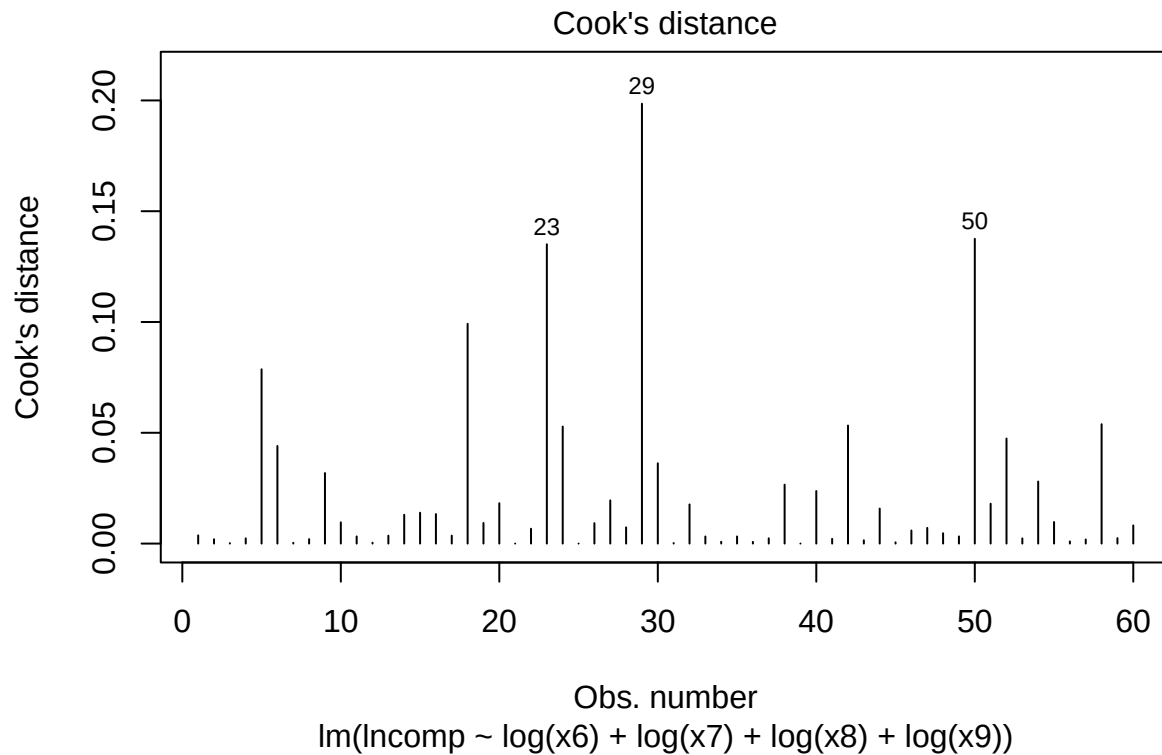
```
plot(log(x9))
points(29,log(x9)[29], col = "red", pch = 16, cex = 2.5)
```



I'd say that only in the $\log(x_9)$, log of the profit, do we see index 29 being particularly outlier-ish.

Question 8

```
plot(lmmodel, which = 4)
```



We see that the 29th index is indeed the most influential, but according to our general rule in class, it isn't greater than 1, so we don't conclude it has 'high' influence, even though comparatively it has the highest influence.

Question 9

a)

let's start with the full model from question 3:

```
summary(lnmodel)
```

```
##
## Call:
## lm(formula = lncomp ~ log(x6) + log(x7) + log(x8) + log(x9))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.77458 -0.33331 -0.07043  0.29030  1.19390
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.49990    0.51439  10.692 4.81e-15 ***
## log(x6)        0.09903    0.08656   1.144  0.25755
## log(x7)        0.29297    0.10152   2.886  0.00557 **
## log(x8)       -0.24594    0.10592  -2.322  0.02397 *
## log(x9)       -0.06740    0.10216  -0.660  0.51217
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.4585 on 55 degrees of freedom
## Multiple R-squared: 0.3644, Adjusted R-squared: 0.3182
## F-statistic: 7.884 on 4 and 55 DF, p-value: 4.261e-05
```

Since $\log(x_9)$ has the highest p value that is also above 0.15, we drop it:

```
droppedx9 <- lm(lncomp ~ log(x6)+log(x7)+log(x8))
summary(droppedx9)
```

```
##
## Call:
## lm(formula = lncomp ~ log(x6) + log(x7) + log(x8))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.78287 -0.31337 -0.08203  0.30202  1.16428
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.42552    0.49934  10.865 2.07e-15 ***
## log(x6)       0.08310    0.08270   1.005 0.31931
## log(x7)       0.25316    0.08123   3.117 0.00289 **
## log(x8)      -0.20022    0.07971  -2.512 0.01491 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4562 on 56 degrees of freedom
## Multiple R-squared: 0.3594, Adjusted R-squared: 0.3251
## F-statistic: 10.47 on 3 and 56 DF, p-value: 1.436e-05
```

Since $\log(x_6)$ has the highest p value above 0.15, we drop it:

```
droppedx9x6 <- lm(lncomp ~ log(x7)+log(x8))
summary(droppedx9x6)
```

```
##
## Call:
## lm(formula = lncomp ~ log(x7) + log(x8))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.77616 -0.32415 -0.06035  0.35129  1.11142
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.89419    0.17832  33.053 < 2e-16 ***
## log(x7)       0.31175    0.05656   5.512 8.94e-07 ***
## log(x8)      -0.25792    0.05528  -4.665 1.91e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4563 on 57 degrees of freedom
## Multiple R-squared: 0.3478, Adjusted R-squared: 0.325
## F-statistic: 15.2 on 2 and 57 DF, p-value: 5.119e-06
```

Since no other explanatory variate is above 0.15, we keep this as our model.

b)

```
lnX <- as.matrix(cbind(log(x6),log(x7),log(x8),log(x9)))
cp_out <- leaps(x = lnX, y = lncomp, method = "Cp", nbest = 1)
adjr2_out <- leaps(x = lnX, y = lncomp, method = "adjr2", nbest = 1)
cp_out$Cp
```

```
## [1] 10.237745 2.434732 3.435257 5.000000
```

```
cp_out$which
```

```
##      1      2      3      4
## 1  TRUE FALSE FALSE FALSE
## 2 FALSE  TRUE  TRUE FALSE
## 3  TRUE  TRUE  TRUE FALSE
## 4  TRUE  TRUE  TRUE  TRUE
```

```
adjr2_out$adjr2
```

```
## [1] 0.2213539 0.3249527 0.3250671 0.3181912
```

We pick $\log(x6)$, $\log(x7)$ and $\log(x8)$, bc its the highest adjR^2 (barely though) that fits our mallows cp of being strictly less than $k+1$.

```
pickedmodel <- lm(lncomp ~ log(x6)+log(x7)+log(x8))
```

c)

```
n <- length(lncomp)
mse_full <- summary(lmmodel)$sigma^2
sse_reduced <- sum(resid(pickedmodel)^2)
k <- 3+1

Cp_check <- sse_reduced/mse_full - (n - 2*k)
Cp_check
```

```
## [1] 3.435257
```

Which is the same mallows cp as expected.

d)

```
summary(pickedmodel)
```

```
##
## Call:
## lm(formula = lncomp ~ log(x6) + log(x7) + log(x8))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.78287 -0.31337 -0.08203  0.30202  1.16428
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.42552    0.49934  10.865 2.07e-15 ***
## log(x6)        0.08310    0.08270   1.005  0.31931
## log(x7)        0.25316    0.08123   3.117  0.00289 **
```

```
## log(x8)      -0.20022    0.07971  -2.512  0.01491 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4562 on 56 degrees of freedom
## Multiple R-squared:  0.3594, Adjusted R-squared:  0.3251
## F-statistic: 10.47 on 3 and 56 DF,  p-value: 1.436e-05
```

```
summary(droppedx9x6)
```

```
##
## Call:
## lm(formula = lncomp ~ log(x7) + log(x8))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.77616 -0.32415 -0.06035  0.35129  1.11142
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.89419    0.17832  33.053 < 2e-16 ***
## log(x7)      0.31175    0.05656   5.512 8.94e-07 ***
## log(x8)     -0.25792    0.05528  -4.665 1.91e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4563 on 57 degrees of freedom
## Multiple R-squared:  0.3478, Adjusted R-squared:  0.325
## F-statistic: 15.2 on 2 and 57 DF,  p-value: 5.119e-06
```

they are slightly different, but honestly its the exact same comparision made with the leaps function, with both models fitting our mallows cp criteria. Our picked model has 0.0001 more adj R^2 value, which is basically a negligible difference in my opinion.

e)

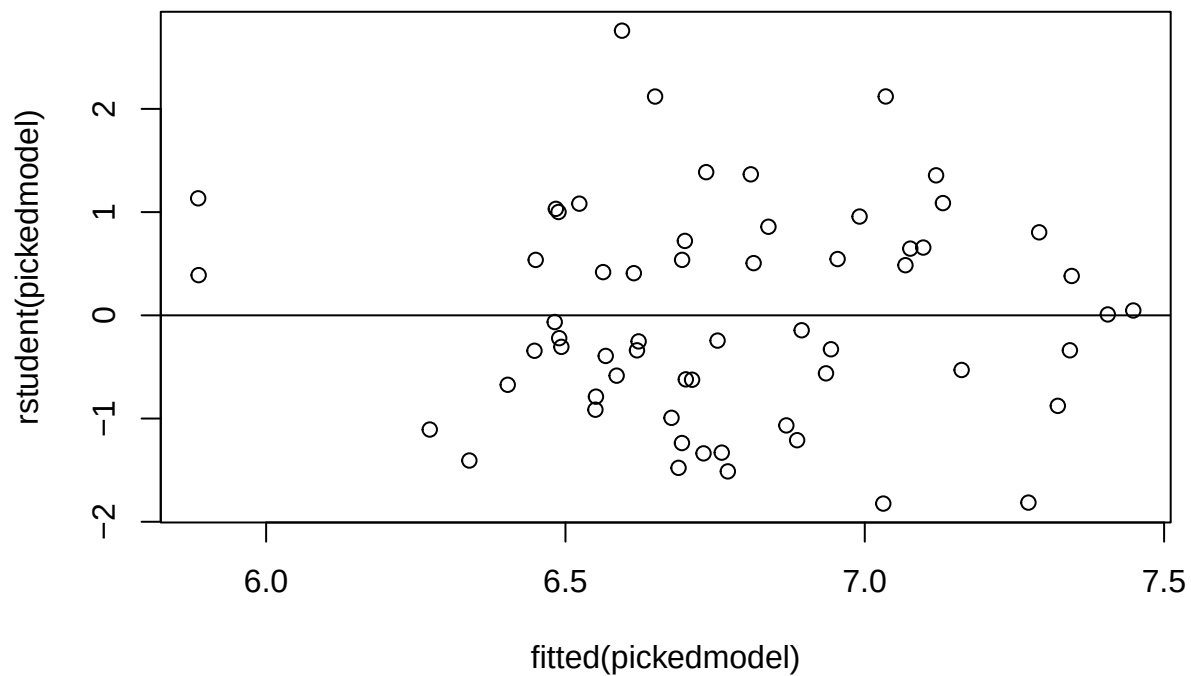
```
anova(pickedmodel,lnmodel)
```

```
## Analysis of Variance Table
##
## Model 1: lncomp ~ log(x6) + log(x7) + log(x8)
## Model 2: lncomp ~ log(x6) + log(x7) + log(x8) + log(x9)
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      56 11.656
## 2      55 11.564  1  0.091517 0.4353 0.5122
```

From the test, we cant really say with confidence the dropped log(x9) variate, the log of the profit of the company, contributed significantly to the log of the CEO compensation.

f)

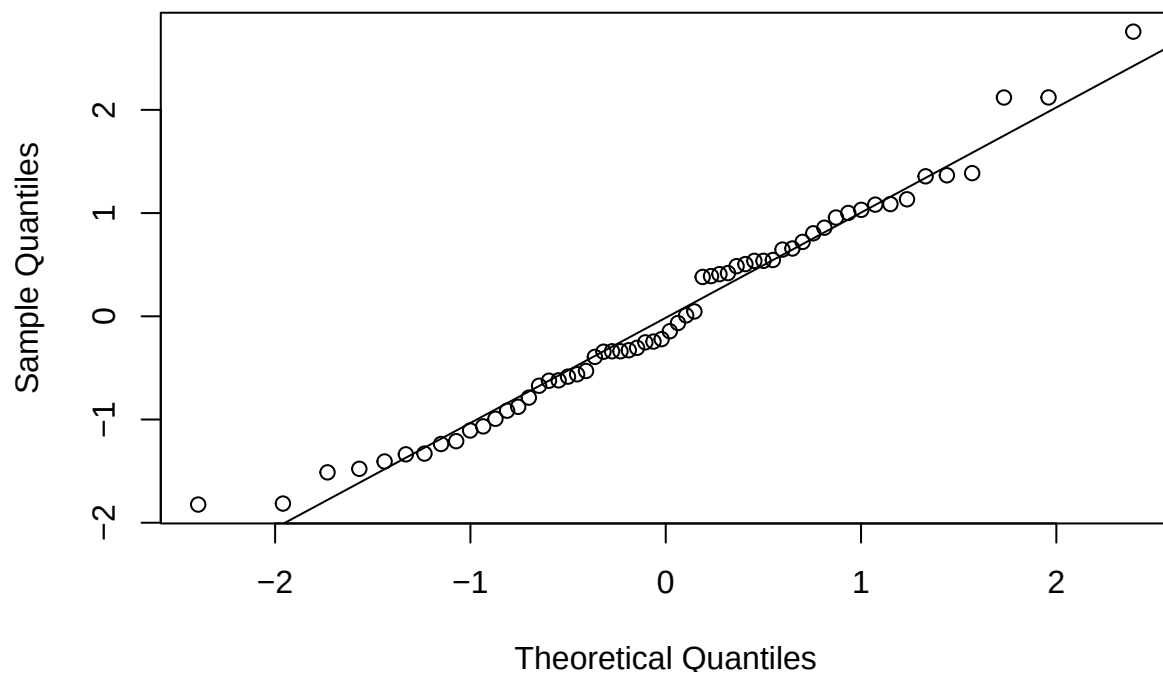
```
plot(fitted(pickedmodel), rstudent(pickedmodel))
abline(h = 0)
```



Once again, very similar res vs fitted values as previous.

```
qqnorm(rstudent(pickedmodel)); qqline(rstudent(pickedmodel))
```

Normal Q-Q Plot



Very close to normal distribution, still with a little light of tails, though.

g)

```
new_lnx <- data.frame(x6=3100,x7=2.1,x8=1.1)
exp(predict(pickedmodel,new_lnx,interval='prediction',level=.95))
```

```
##          fit          lwr          upr
## 1 524.4452 190.5552 1443.376
```

with a compensation of 1.5 million, we can now conclude that the ceo is being paid a little too much, given our model, and what we expected. As for the interval, it is tighter than the previous, as shown here:

```
new_x <- data.frame(x1=48,x2=1,x4=14,x5=3,x6=3100,x7=2.1,x8=1.1,x9=172)
predict(redmodel,new_x,interval='prediction',level=.95)
```

```
##          fit          lwr          upr
## 1 761.5257 -268.8854 1791.937
```

So we went from a width of around 2000, to a width of around 1200, which is certainly a increase in certainty.